SHUBHAM JOSHI

Washington, DC
(202) 440-8445
shubhamjoshipro.mail@gmail.com
LinkedIn
GitHub
Portfolio

CORE SKILLS

AI Systems & GenAI
- RAG pipelines
- LLM evaluation
- Agentic systems
- Prompt engineering
- Claude & OpenAI APIs
- Schema validation

Machine Learning
- LightGBM, XGBoost
- Logistic Regression
- Model evaluation
- Cross-validation
- Calibration, PSI
- SHAP, LIME

Data & Analytics
- SQL (PostgreSQL, CTEs, window functions)
- Python (Pandas, NumPy)
- ETL pipelines
- Statistical analysis
- A/B testing
- Clustering

Engineering
- FastAPI
- Asyncio
- REST APIs
- CI/CD (Railway)
- System design

FOCUS AREAS

- LLM boundary design
- AI evaluation & failure modes

PROFESSIONAL PROFILE

I build and evaluate AI systems at the intersection of machine learning, decision-making, and real-world deployment. My work spans clinical decision support (RxCheck), consumer AI systems (FridgeChef), and context-aware agent design (Nimbus), with a focus on systems that are interpretable, auditable, and aligned with real-world constraints.

Alongside system design, I work extensively with data, building pipelines, conducting statistical analysis, and translating large-scale datasets into decision-making insights.

A central theme across my work is identifying and mitigating failure modes in AI systems, including hallucinations, missing evidence, and inconsistent reasoning under constraint.

DATA & ANALYTICS EXPERIENCE

- Built ETL pipelines integrating CMS datasets (695M+ enrollee-months, 138K+ plans)
- Applied clustering and statistical analysis to identify national vs nonprofit strategy differences
- Conducted SQL/Python-based analysis (aggregation, cohort analysis, feature engineering)
- Performed A/B testing and statistical evaluation in consulting contexts
- Translated raw data into actionable insights for real-world decision-making

PROJECTS & SYSTEMS

RxCheck: *Clinical Decision Support & LLM Boundary System*
- Designed a three-layer architecture separating deterministic logic from LLM explanation layer
- Restricted LLM to RAG-based explanation (no decision authority)
- Implemented schema validation and failure logging for all model outputs
- Built RxNorm normalization pipeline (exact → fuzzy → NDC fallback)
- Designed audit systems (snapshots, prompt versioning, override logs)

FridgeChef: *Multi-Provider AI System*
- Built multi-provider LLM system optimizing cost vs quality (Claude Vision vs GPT-4o-mini)
- Engineered real-time streaming pipeline (thread→async bridge, SSE keepalive)
- Designed evaluation framework tracking output quality, cost, and reliability
- Implemented multi-source ingestion and fuzzy data deduplication

Nimbus: *Context-Aware AI Companion*
- Built context-driven system using structured behavioral data (sleep, workload, deadlines)
- Designed agent workflows for prioritization, recovery, and communication
- Focused on reducing cognitive load and improving decision-making

Healthcare Policy Data Pipeline (CMS)
- Built ETL pipeline across large-scale healthcare datasets
- Applied clustering and statistical analysis to identify policy trends

- Data-driven decision systems
- Human-aware AI systems
- Cost-aware AI engineering

ADDITIONAL

- PyTorch (coursework)
- RL concepts (MDPs, RLHF)
- Transformer architectures
- LLM experimentation

EDUCATION

MS Business Analytics & AI
Johns Hopkins University
Dean's Scholarship
Expected Aug 2026

BBA Global Business
NUCB Japan

- Produced insights used in real-world advocacy strategy

Home Credit Default Risk: *ML Evaluation, 7th in Kaggle Hackathon*

- Built ML pipeline (LightGBM, cross-validation, class imbalance handling)
- Conducted SHAP/LIME explainability and fairness analysis
- Implemented calibration, drift detection, and threshold optimization

PROFESSIONAL EXPERIENCE

VP & Co-Founder | Business Analytics & AI Think Tank
Johns Hopkins University **2025–Present**

- Lead research on AI systems, evaluation, and deployment (3×/week)
- Designed and led workshops deploying ML systems with explainability
- Leadership Position responsible for ~100 members

Independent Data & Analytics Consultant
US / Japan / Australia **2021–2025**

- Delivered analytics, experimentation, and strategy solutions
- Conducted A/B testing, statistical analysis, and decision support